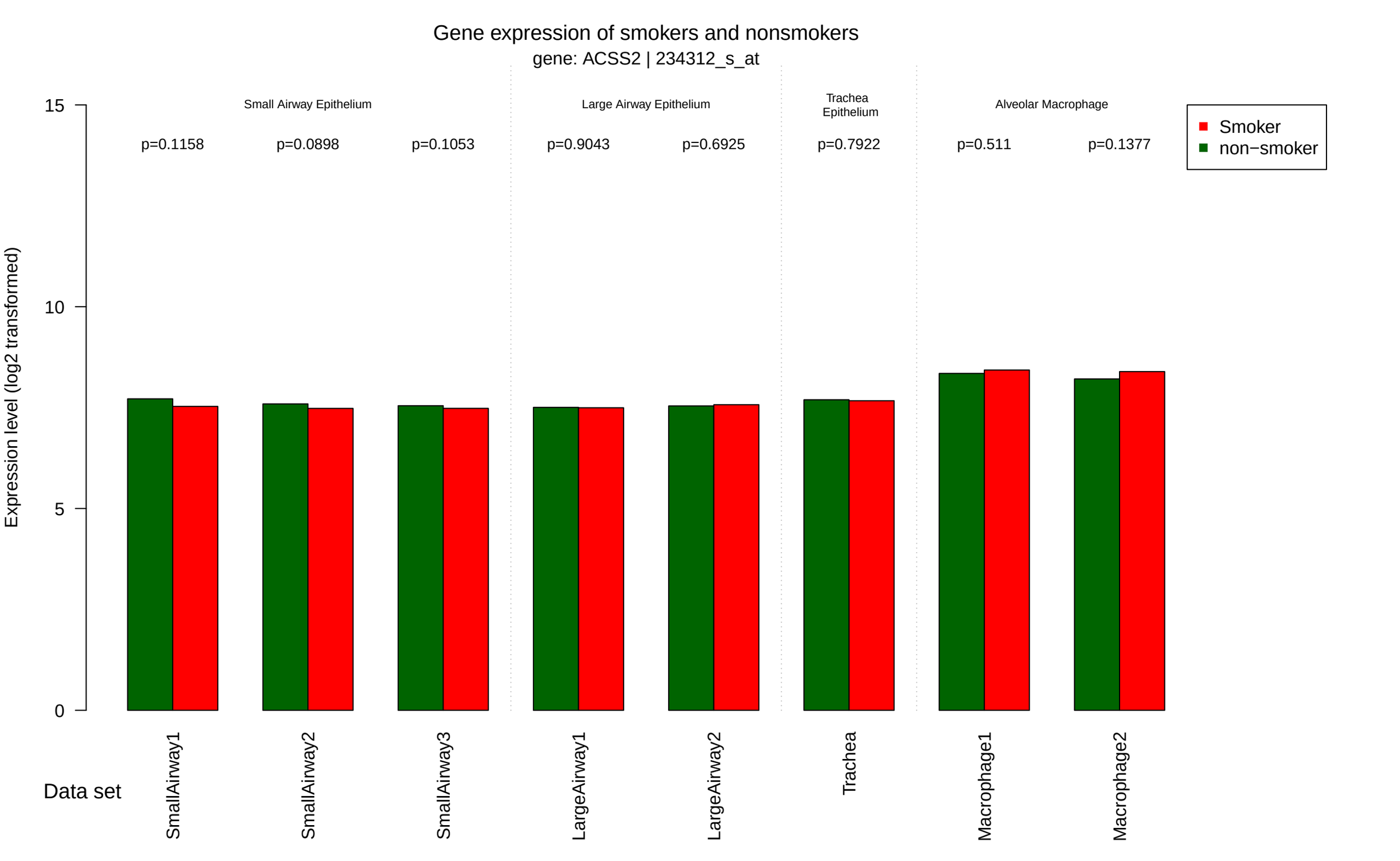
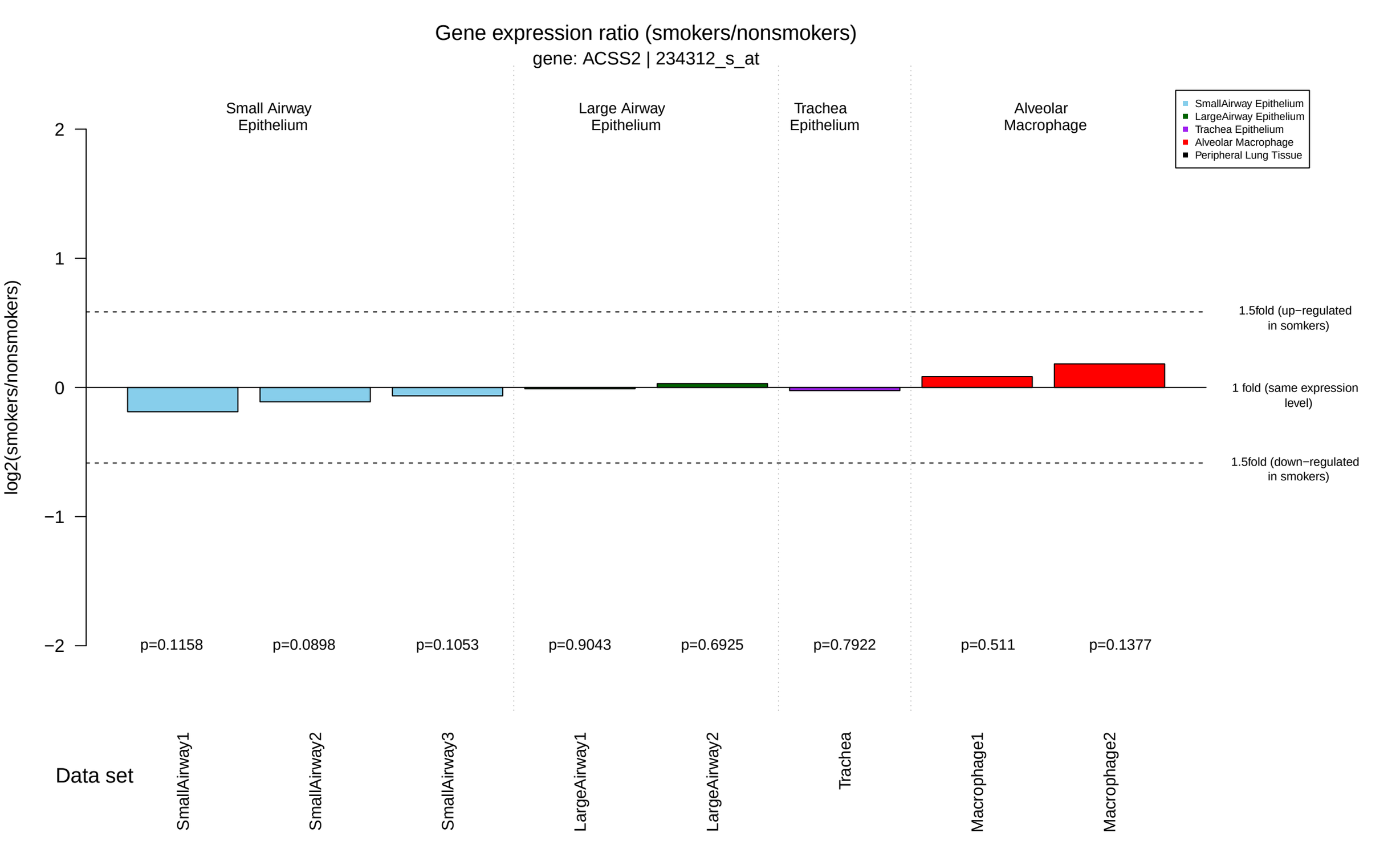
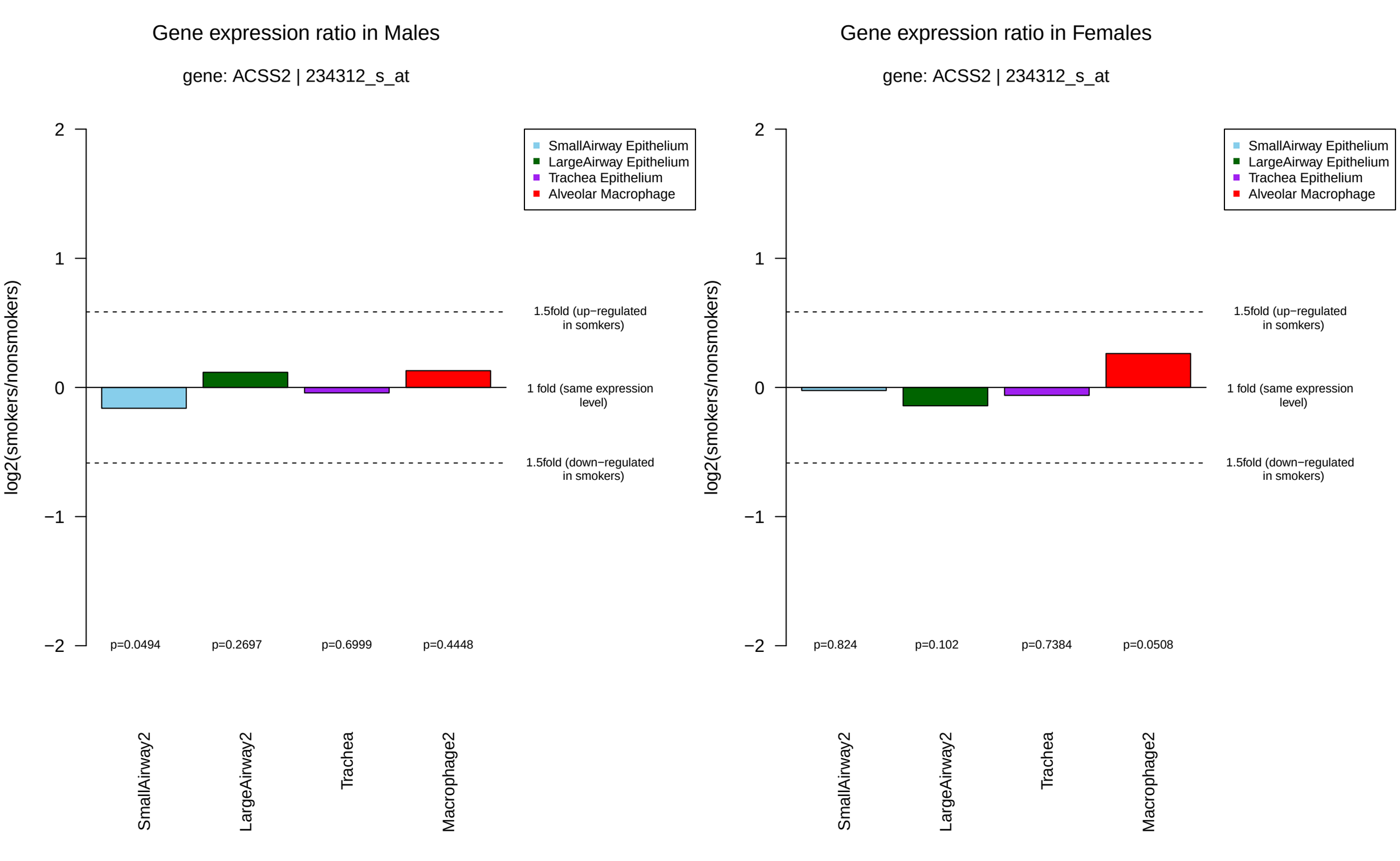
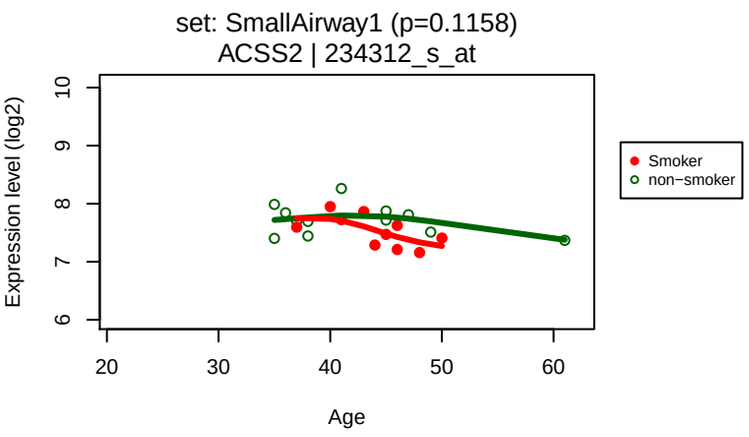
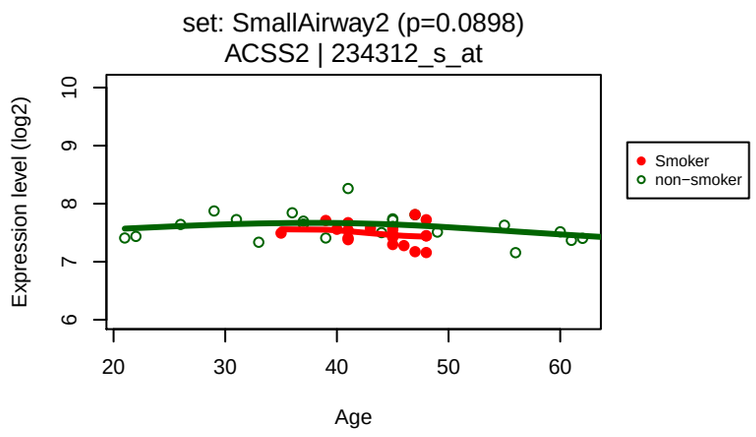
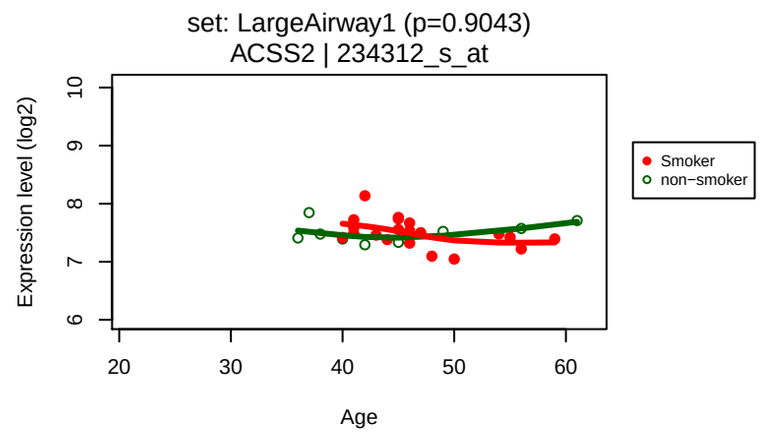

Gene expression of smokers and nonsmokers
gene: ACSS2 | 234312_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1158
p=0.0898
p=0.1053
p=0.9043
p=0.6925
p=0.7922
p=0.511
p=0.1377
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

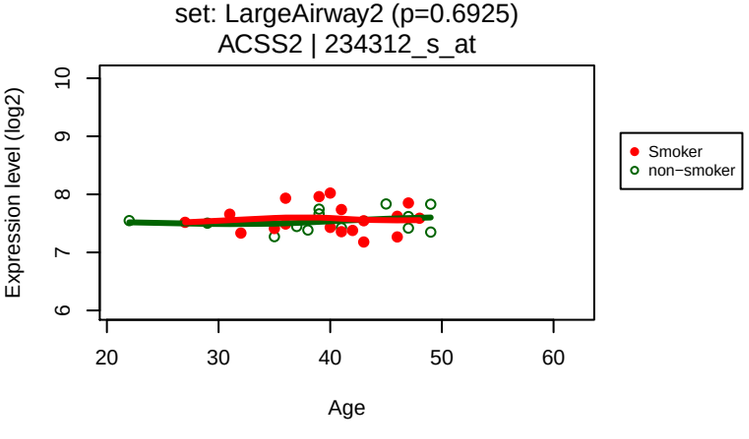

Gene expression ratio (smokers/nonsmokers)
gene: ACSS2 | 234312_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1158
p=0.0898
p=0.1053
p=0.9043
p=0.6925
p=0.7922
p=0.511
p=0.1377
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

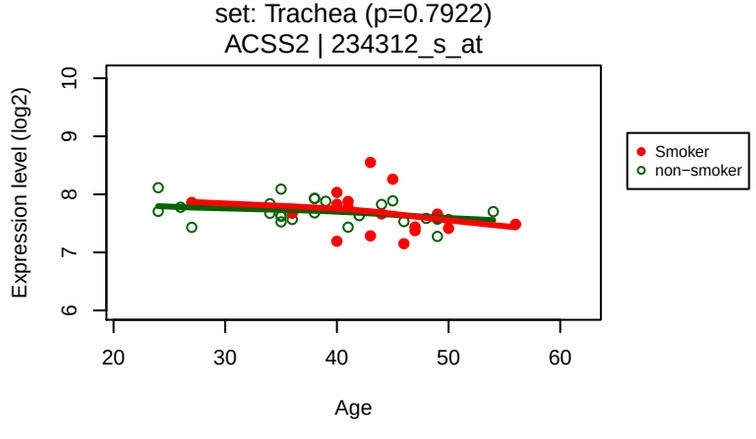

Gene expression ratio in Males
gene: ACSS2 | 234312_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0494
p=0.2697
p=0.6999
p=0.4448
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACSS2 | 234312_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.824
p=0.102
p=0.7384
p=0.0508
SmallAirway2
LargeAirway2
Trachea
Macrophage2

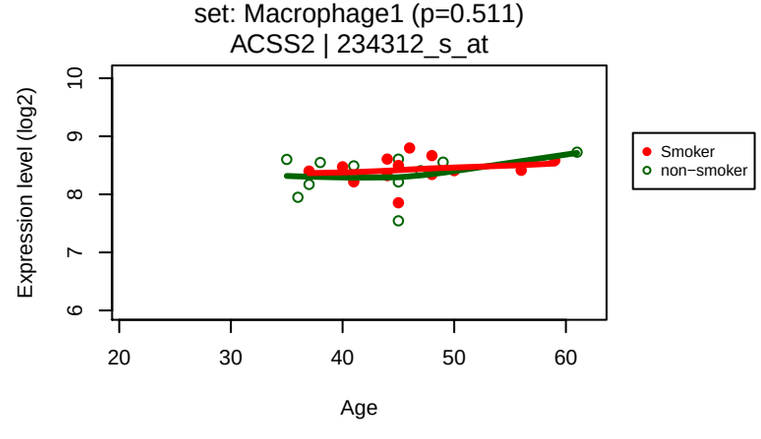

set: SmallAirway1 (p=0.1158)
ACSS2 | 234312_s_at
Expression level (log2)
Age
Smoker
non−smoker

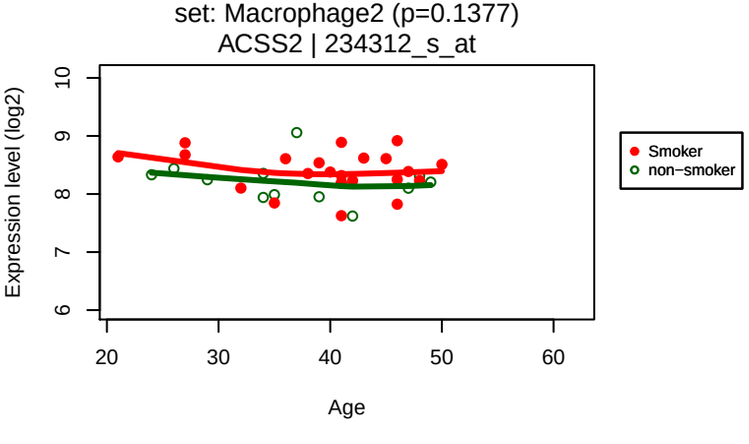

set: SmallAirway2 (p=0.0898)
ACSS2 | 234312_s_at
Expression level (log2)
Age
Smoker
non−smoker

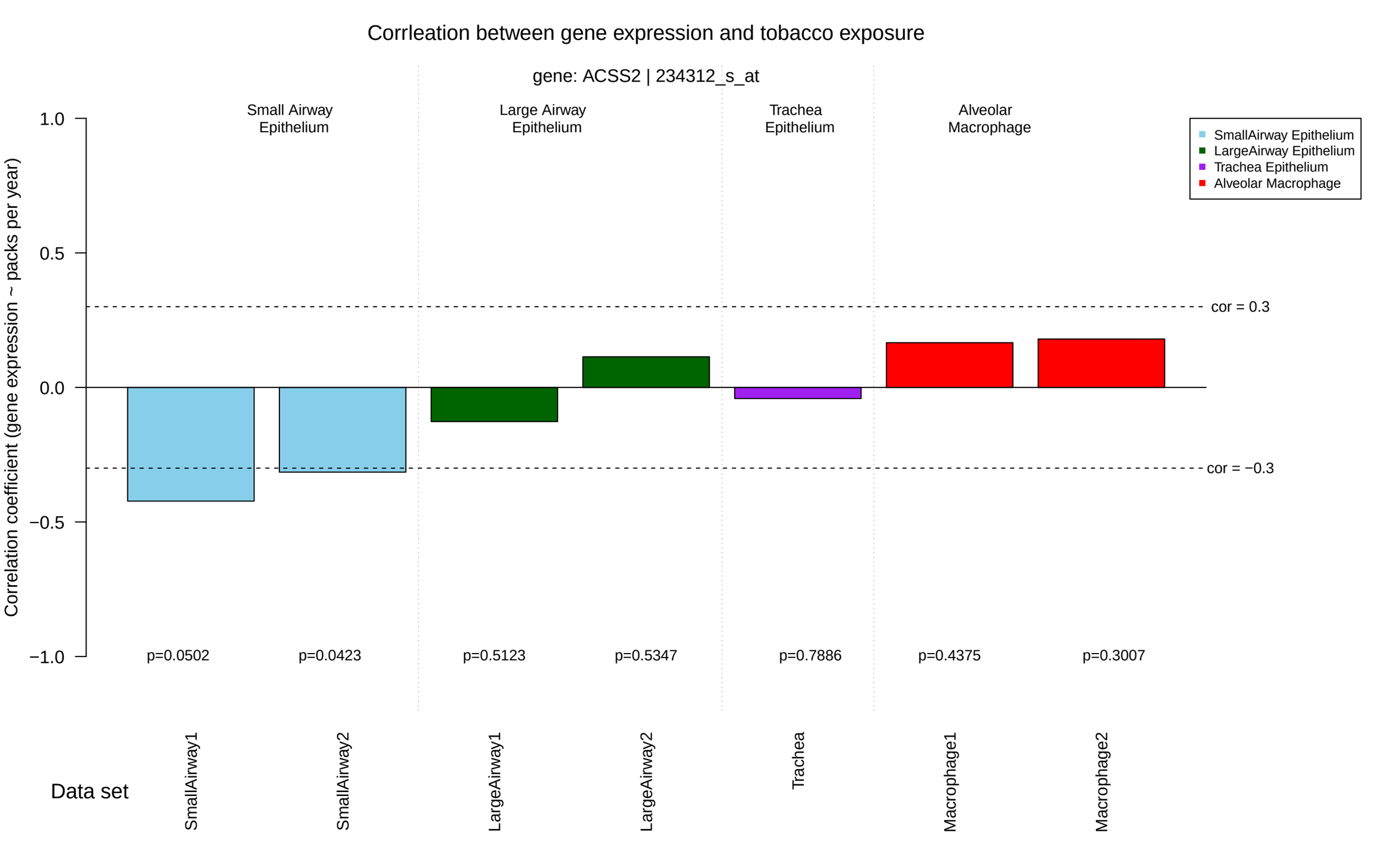

set: LargeAirway1 (p=0.9043)
ACSS2 | 234312_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6925)
ACSS2 | 234312_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7922)
ACSS2 | 234312_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.511)
ACSS2 | 234312_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1377)
ACSS2 | 234312_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACSS2 | 234312_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0502
p=0.0423
p=0.5123
p=0.5347
p=0.7886
p=0.4375
p=0.3007
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2